

## § 5—1

1. 利用定积分的几何意义,

填写下列定积分值:

$$(1) \int_0^{2\pi} \cos x dx = \underline{0}.$$

解: 原式 = 0.

$$(2) \int_0^2 \sqrt{1 - \frac{1}{4}x^2} dx = \underline{\frac{\pi}{2}}.$$

$$\text{解: 原式} = \int_0^2 \sqrt{\frac{4-x^2}{4}} dx$$

$$= \frac{1}{2} \int_0^2 \sqrt{4-x^2} dx$$

$$= \frac{1}{2} \cdot \frac{1}{4} \pi \cdot 2^2 = \frac{\pi}{2}.$$

上式表示介于  $x=0$ ,  $x=2$ ,  $y=0$

和  $y = \sqrt{4-x^2}$  之间的平面图形

( $\frac{1}{4}$  圆) 的面积的一半.

(或解为:  $\int_0^2 \sqrt{1 - \frac{1}{4}x^2} dx$  在几何上

表示椭圆  $\frac{x^2}{4} + y^2 \leq 1$  在第一象限内

的面积, 即  $\frac{1 \times 2\pi}{4} = \frac{\pi}{2}$ .)

2. 在积分中值定理中,

设函数  $f(x)$  在  $[a, b]$  上连续,

则至少存在一点  $\xi \in [a, b]$ ,

$$\text{使得 } f(\xi) = \underline{\frac{1}{b-a} \int_a^b f(x) dx}.$$

3. 由定积分的性质,

比较积分值的大小:

$$\int_0^1 \frac{x}{1+x} dx \text{ 与 } \int_0^1 \ln(1+x) dx.$$

解: 令  $F(x) = \ln(1+x) - \frac{x}{1+x}$ , 则

$$F'(x) = \frac{1}{1+x} - \frac{1}{(1+x)^2} = \frac{x}{(1+x)^2} > 0,$$

$$x \in (0,1).$$

由于  $F(x)$  在闭区间  $[0,1]$  上连续,

所以  $F(x)$  在  $[0,1]$  上单调增加, 故

$$F(x) > F(0) = 0, \quad x \in (0,1].$$

$$\text{即 } \ln(1+x) > \frac{x}{1+x}, \quad x \in (0,1].$$

$$\text{所以当 } 0 \leq x \leq 1 \text{ 时, } \frac{x}{1+x} \leq \ln(1+x),$$

(课本 P129 例)

此处, 等号仅当  $x=0$  时成立.

$$\text{因为在 } [0,1] \text{ 上, } \frac{x}{1+x} \neq \ln(1+x).$$

根据定积分的性质及课本习题

[P237 12(1)] 的结论, 所以

$$\int_0^1 \frac{x}{1+x} dx < \int_0^1 \ln(1+x) dx.$$

4. 利用估值定理证明:

$$\frac{\pi}{9} \leq \int_{\frac{\sqrt{3}}{3}}^{\sqrt{3}} x \arctan x dx \leq \frac{2\pi}{3}.$$

证明: 设  $f(x) = x \arctan x$ ,

$$x \in \left[ \frac{\sqrt{3}}{3}, \sqrt{3} \right].$$

$$\text{则 } f'(x) = \arctan x + \frac{x}{1+x^2} > 0,$$

$$x \in \left[ \frac{\sqrt{3}}{3}, \sqrt{3} \right], \text{ 所以}$$

$$f(x) = x \arctan x \text{ 在 } \left[ \frac{\sqrt{3}}{3}, \sqrt{3} \right] \text{ 上单调增加,}$$

因此  $f(x)$  在闭区间  $\left[\frac{\sqrt{3}}{3}, \sqrt{3}\right]$  上的

最小值和最大值分别为

$$f\left(\frac{\sqrt{3}}{3}\right) = \frac{\sqrt{3}}{3} \arctan \frac{\sqrt{3}}{3} = \frac{\sqrt{3}}{3} \cdot \frac{\pi}{6} = \frac{\sqrt{3}\pi}{18},$$

$$f(\sqrt{3}) = \sqrt{3} \arctan \sqrt{3} = \sqrt{3} \cdot \frac{\pi}{3} = \frac{\sqrt{3}\pi}{3}.$$

故  $\frac{\sqrt{3}\pi}{18} \leq x \arctan x \leq \frac{\sqrt{3}\pi}{3}.$

由定积分的估值性质, 得

$$\begin{aligned} \frac{\pi}{9} &= \frac{\sqrt{3}\pi}{18} \left( \sqrt{3} - \frac{\sqrt{3}}{3} \right) \\ &\leq \int_{\frac{\sqrt{3}}{3}}^{\sqrt{3}} x \arctan x dx \\ &\leq \frac{\sqrt{3}\pi}{3} \left( \sqrt{3} - \frac{\sqrt{3}}{3} \right) = \frac{2\pi}{3}, \end{aligned}$$

所以  $\frac{\pi}{9} \leq \int_{\frac{\sqrt{3}}{3}}^{\sqrt{3}} x \arctan x dx \leq \frac{2\pi}{3}.$

## § 5—2

1. 设  $f(x)$  连续,  $F'(x) = f(x)$ ,

$C$  为任意常数, 则正确的选项  
为 (A, B, C, E).

(A)  $\int f(x) dx = \int_a^x f(x) dx + C;$

(B)  $\int f(x) dx = C - \int_x^b f(x) dx;$

(C)  $\int_a^b f(t) dt = \int_a^b f(u) du;$

(D)  $\int_a^x F'(x) dx = F(x);$

(E)  $\frac{d}{dx} \int_a^x f(t) dt = \frac{d}{dx} \int f(x) dx;$

$$(F) \int F'(x)dx = F(x) + C.$$

$$\text{解: (A)} \int f(x)dx = F(x) + C,$$

$$\begin{aligned} \int_a^x f(x)dx + C &= [F(x)]_a^x + C \\ &= F(x) - F(a) + C. \end{aligned}$$

$$\begin{aligned} (B) \quad C - \int_x^b f(x)dx &= C - F(b) + F(x) \\ &= F(x) + C'. \end{aligned}$$

2. 计算下列各导数:

$$(1) \frac{d}{dx} \left( \int_0^{x^2} \sqrt{1+t^2} dt \right) = \underline{2x\sqrt{1+x^4}}.$$

$$\text{解: 原式} = \sqrt{1+(x^2)^2} \cdot \frac{dx^2}{dx} = 2x\sqrt{1+x^4}.$$

$$(2) \frac{d}{dx} \left( \int_{x^3}^1 \frac{dt}{\sqrt{1+t^2}} \right) = \underline{-\frac{3x^2}{\sqrt{1+x^6}}}.$$

$$\begin{aligned} \text{解: 原式} &= -\frac{1}{\sqrt{1+(x^3)^2}} \cdot \frac{dx^3}{dx} \\ &= -\frac{3x^2}{\sqrt{1+x^6}}. \end{aligned}$$

$$\begin{aligned} (3) \quad \frac{d}{dx} \left( \int_{\sin x}^{\cos x} e^{t^2} dt \right) &= \\ &= \underline{-e^{\cos^2 x} \sin x - e^{\sin^2 x} \cos x}. \end{aligned}$$

$$\begin{aligned} \text{解: 原式} &= e^{\cos^2 x} (\cos x)' - e^{\sin^2 x} (\sin x)' \\ &= -e^{\cos^2 x} \sin x - e^{\sin^2 x} \cos x. \end{aligned}$$

$$(4) \frac{d}{dx} \left( \int_0^x (x-t) f(t) dt \right) = \underline{\int_0^x f(t) dt}.$$

解: 原式

$$\begin{aligned} &= \frac{d}{dx} \left( x \int_0^x f(t) dt \right) - \frac{d}{dx} \left( \int_0^x t f(t) dt \right) \\ &= \int_0^x f(t) dt + x f(x) - x f(x) \end{aligned}$$

$$= \int_0^x f(t) dt.$$

3. 设  $\begin{cases} x = \int_0^t \frac{\sin u}{u} du, \\ y = \sin t - t \cos t \end{cases}$ , 求  $\left. \frac{d^2 y}{dx^2} \right|_{t=\frac{\pi}{2}}$ .

解:  $\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{\cos t - \cos t + t \sin t}{\frac{\sin t}{t}}$

$$= t^2,$$

$$\frac{d^2 y}{dx^2} = \frac{d\left(\frac{dy}{dx}\right)}{dx} = \frac{d\left(\frac{dy}{dx}\right)}{\frac{dx}{dt}}$$

$$= \frac{2t}{\frac{\sin t}{t}} = \frac{2t^2}{\sin t},$$

所以  $\left. \frac{d^2 y}{dx^2} \right|_{t=\frac{\pi}{2}} = \left. \frac{2t^2}{\sin t} \right|_{t=\frac{\pi}{2}} = \frac{\pi^2}{2}.$

4. 求  $\lim_{x \rightarrow +\infty} \frac{\int_0^x (\arctan t)^2 dt}{\sqrt{x^2 + 1}}.$   $\left( \frac{\infty}{\infty} \right)$

解: 应用洛必达法则, 得

$$\begin{aligned} \text{原式} &= \lim_{x \rightarrow +\infty} \frac{\frac{d}{dx} \left( \int_0^x (\arctan t)^2 dt \right)}{\frac{d}{dx} (\sqrt{x^2 + 1})} \\ &= \lim_{x \rightarrow +\infty} \frac{(\arctan x)^2}{2x} = \lim_{x \rightarrow +\infty} \frac{(\arctan x)^2}{x} \cdot \frac{x}{2\sqrt{x^2 + 1}} \end{aligned}$$

$$= \lim_{x \rightarrow +\infty} \frac{(\arctan x)^2}{\frac{1}{\sqrt{1 + \frac{1}{x^2}}}} = \frac{\left(\frac{\pi}{2}\right)^2}{\sqrt{1 + 0}} = \frac{\pi^2}{4}.$$

5. 若  $f(x)$  为  $(-\infty, +\infty)$  内的连续函数,

求极限  $\lim_{x \rightarrow 0} \frac{\int_0^{x^2} t f(t) dt}{x^4}$ .

解：应用洛必达法则，得

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{\int_0^{x^2} t f(t) dt}{x^4} & \left( \frac{0}{0} \right) \\ &= \lim_{x \rightarrow 0} \frac{x^2 f(x^2) \cdot (x^2)'}{4x^3} = \lim_{x \rightarrow 0} \frac{f(x^2) \cdot 2x}{4x} \\ &= \frac{1}{2} \lim_{x \rightarrow 0} f(x^2) \quad (\text{连续}) \\ &= \frac{1}{2} f(0). \end{aligned}$$

6. 设  $f(x) = \begin{cases} 1, & x \in [0, 1], \\ x, & x \in [1, 2] \end{cases}$ ,

求  $\Phi(x) = \int_0^x f(t) dt$  在  $[0, 2]$  上的表达式，

并讨论  $\Phi(x)$  在  $(0, 2)$  内的连续性.

解：当  $0 \leq x \leq 1$  时，

$$\Phi(x) = \int_0^x f(t) dt = \int_0^x 1 dt = x;$$

当  $1 \leq x \leq 2$  时，

$$\Phi(x) = \int_0^x f(t) dt = \int_0^1 1 dt + \int_1^x t dt$$

$$= 1 + \left[ \frac{1}{2} t^2 \right]_1^x = 1 + \frac{1}{2} x^2 - \frac{1}{2}$$

$$= \frac{1}{2} x^2 + \frac{1}{2}. \quad \text{因此}$$

$$\Phi(x) = \begin{cases} x, & 0 \leq x \leq 1, \\ \frac{1}{2} x^2 + \frac{1}{2}, & 1 < x \leq 2. \end{cases}$$

因为  $\Phi(1) = 1$ ,

$$\lim_{x \rightarrow 1^-} \Phi(x) = \lim_{x \rightarrow 1^-} 1 = 1,$$

$$\lim_{x \rightarrow 1^+} \Phi(x) = \lim_{x \rightarrow 1^+} \left( \frac{1}{2} x^2 + \frac{1}{2} \right) = \frac{1}{2} + \frac{1}{2} = 1,$$

所以  $\Phi(x)$  在  $x = 1$  点连续，

从而在  $(0,2)$  内连续.

5. 证明方程  $3x-1-\int_0^x \frac{1}{1+t^4} dt = 0$

在区间  $(0,1)$  内有且仅有一个实根.

证明: 设  $F(x) = 3x-1-\int_0^x \frac{1}{1+t^4} dt$ ,

则函数  $F(x)$  在闭区间  $[0,1]$  上连续,

且  $F(0) = 0-1-\int_0^0 \frac{1}{1+t^4} dt = -1 < 0$ ,

$$\begin{aligned} F(1) &= 3-1-\int_0^1 \frac{1}{1+t^4} dt \\ &= 2-\int_0^1 \frac{1}{1+t^4} dt > 0, \end{aligned}$$

由零点定理 (介值定理) 知

函数  $F(x)$  在  $(0,1)$  内有至少有一零点.

又  $F'(x) = 3-\frac{1}{1+x^4} > 0, x \in (0,1)$ .

则函数  $F(x)$  在闭区间  $[0,1]$  上单调增加,

从而零点唯一, 所以原方程

在区间  $(0,1)$  内有且仅有一个实根.

### § 5—3

#### 1. 填空题

(1)  $\int_{-3}^3 (x-3)\sqrt{9-x^2} dx = -\frac{27}{2}\pi$ .

解: 原式

$$= \int_{-3}^3 x\sqrt{9-x^2} dx - 3 \int_{-3}^3 \sqrt{9-x^2} dx$$

(奇函数) (偶函数)

$$= 0 - 6 \int_0^3 \sqrt{9-x^2} dx$$

$$= -6 \cdot \frac{1}{4} \cdot \pi \cdot 3^2 = -\frac{27}{2}\pi.$$

(2)  $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \left[ \frac{(1+x)\cos x}{1+\sin^2 x} + \ln(x+\sqrt{1+x^2}) \right] dx$

$$= \frac{\pi}{2}.$$

解：原式

$$= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\cos x}{1 + \sin^2 x} dx \quad (\text{偶函数})$$

$$+ \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{x \cos x}{1 + \sin^2 x} dx \quad (\text{奇函数})$$

$$+ \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \ln(x + \sqrt{1+x^2}) dx \quad (\text{奇函数})$$

$$= 2 \int_0^{\frac{\pi}{2}} \frac{\cos x}{1 + \sin^2 x} dx + 0 + 0$$

$$= 2 \int_0^{\frac{\pi}{2}} \frac{1}{1 + \sin^2 x} d(\sin x)$$

$$= 2 [\arctan \sin x]_0^{\frac{\pi}{2}} = 2 \left( \frac{\pi}{4} - 0 \right)$$

$$= \frac{\pi}{2}.$$

2. 计算下列定积分：

$$(1) \int_{\frac{1}{\sqrt{2}}}^1 \frac{\sqrt{1-x^2}}{x^2} dx.$$

$$\text{原式} = - \int_{\frac{1}{\sqrt{2}}}^1 \sqrt{1-x^2} d\left(\frac{1}{x}\right)$$

$$= - \left[ \frac{1}{x} \sqrt{1-x^2} \right]_{\frac{1}{\sqrt{2}}}^1 + \int_{\frac{1}{\sqrt{2}}}^1 \frac{1}{x} d(\sqrt{1-x^2})$$

$$= 1 - \int_{\frac{1}{\sqrt{2}}}^1 \frac{2x}{2\sqrt{1-x^2}} \cdot \frac{1}{x} dx$$

$$= 1 - \int_{\frac{1}{\sqrt{2}}}^1 \frac{1}{\sqrt{1-x^2}} dx$$

$$= 1 - [\arcsin x]_{\frac{1}{\sqrt{2}}}^1$$

$$= 1 - \arcsin 1 + \arcsin\left(\frac{1}{\sqrt{2}}\right)$$



$$= 1 - \frac{\pi}{2} + \frac{\pi}{4} = 1 - \frac{\pi}{4}.$$

$$(2) \int_0^3 \frac{x dx}{1 + \sqrt{1+x}}.$$

解: 令  $t = \sqrt{1+x}$ , 则  $x = t^2 - 1$ ,

$$dx = 2t dt.$$

当  $x = 0$  时,  $t = 1$ ; 当  $x = 3$  时,  $t = 2$ .

$$\begin{aligned} \text{于是 原式} &= \int_1^2 \frac{t^2 - 1}{1 + t} \cdot 2t dt \\ &= \int_1^2 (t - 1) \cdot 2t dt = 2 \int_1^2 (t^2 - t) dt \\ &= 2 \left[ \frac{t^3}{3} - \frac{t^2}{2} \right]_1^2 \\ &= 4 \left[ \frac{1}{3}(8 - 1) - \frac{1}{2}(4 - 1) \right] = \frac{5}{3}. \end{aligned}$$

$$(3) \int_0^{\pi} \sqrt{\sin x - \sin^3 x} dx.$$

解: 原式  $= \int_0^{\pi} \sqrt{\sin x (1 - \sin^2 x)} dx$

$$\begin{aligned} &= \int_0^{\pi} |\cos x| \sqrt{\sin x} dx \\ &= \int_0^{\frac{\pi}{2}} \cos x \sqrt{\sin x} dx - \int_{\frac{\pi}{2}}^{\pi} \cos x \sqrt{\sin x} dx \\ &= \int_0^{\frac{\pi}{2}} \sqrt{\sin x} d(\sin x) - \int_{\frac{\pi}{2}}^{\pi} \sqrt{\sin x} d(\sin x) \\ &= \frac{2}{3} \left[ (\sin x)^{\frac{3}{2}} \right]_0^{\frac{\pi}{2}} - \frac{2}{3} \left[ (\sin x)^{\frac{3}{2}} \right]_{\frac{\pi}{2}}^{\pi} \\ &= \frac{2}{3}(1 - 0) - \frac{2}{3}(0 - 1) = \frac{4}{3}. \end{aligned}$$

$$(4) \int_{\frac{1}{e}}^e |\ln x| dx.$$

解: 原式  $= -\int_{\frac{1}{e}}^1 \ln x dx + \int_1^e \ln x dx$

$$= (x \ln x) \Big|_1^e - \int_1^e dx - (x \ln x) \Big|_{\frac{1}{e}}^1 + \int_{\frac{1}{e}}^1 dx$$

$$= e - e + 1 - \frac{1}{e} + 1 - \frac{1}{e} = 2 - \frac{2}{e}.$$

(5) 求  $\int_0^{\frac{\pi^2}{4}} (\sin \sqrt{x})^2 dx$ .

解: 令  $\sqrt{x} = t$ , 则  $x = t^2$ ,  $dx = 2t dt$ .

当  $x = 0$  时,  $t = 0$ ;

当  $x = \frac{\pi^2}{4}$  时,  $t = \frac{\pi}{2}$ . 于是

$$\text{原式} = 2 \int_0^{\frac{\pi}{2}} t (\sin t)^2 dt$$

$$= 2 \int_0^{\frac{\pi}{2}} t \cdot \frac{1 - \cos 2t}{2} dt$$

$$= \int_0^{\frac{\pi}{2}} (t - t \cos 2t) dt$$

$$= \int_0^{\frac{\pi}{2}} t dt - \int_0^{\frac{\pi}{2}} t \cos 2t dt$$

$$= \left[ \frac{t^2}{2} \right]_0^{\frac{\pi}{2}} - \frac{1}{2} \int_0^{\frac{\pi}{2}} t d(\sin 2t)$$

$$= \frac{\pi^2}{8} - \frac{1}{2} [t \sin 2t]_0^{\frac{\pi}{2}} + \frac{1}{2} \int_0^{\frac{\pi}{2}} \sin 2t dt$$

$$= \frac{\pi^2}{8} - 0 - \frac{1}{4} [\cos 2t]_0^{\frac{\pi}{2}}$$

$$= \frac{\pi^2}{8} - \frac{1}{4} [\cos \pi - \cos 0] = \frac{\pi^2}{8} + \frac{1}{2}.$$

3. 设  $f(x) = \begin{cases} e^{-x}, & x \geq 0 \\ 1+x^2, & x < 0 \end{cases}$ ,

求  $\int_1^3 f(x-2) dx$ .

解: 令  $t = x - 2$ , 则  $x = 2 + t$ ,  $dx = dt$ .

当  $x=1$  时,  $t=-1$ ; 当  $x=3$  时,  $t=1$ .

$$\text{于是 } \int_1^3 f(x-2)dx = \int_{-1}^1 f(t)dt$$

$$= \int_{-1}^0 (1+t^2)dt + \int_0^1 e^{-t}dt$$

$$= \left[ t + \frac{t^3}{3} \right]_{-1}^0 - [e^{-t}]_0^1$$

$$= \left[ 0 - \left( -1 - \frac{1}{3} \right) \right] - (e^{-1} - 1)$$

$$= \frac{4}{3} - e^{-1} + 1 = \frac{7}{3} - \frac{1}{e}.$$

$$\text{解法二: 已知 } f(x) = \begin{cases} e^{-x}, & x \geq 0 \\ 1+x^2, & x < 0 \end{cases}.$$

$$\text{则 } f(x-2) = \begin{cases} e^{-(x-2)}, & x \geq 2 \\ 1+(x-2)^2, & x < 2 \end{cases},$$

$$\text{于是 } \int_1^3 f(x-2)dx$$

$$= \int_1^2 f(x-2)dx + \int_2^3 f(x-2)dx$$

$$= \int_1^2 [1+(x-2)^2]dx + \int_2^3 e^{-(x-2)}dx$$

$$= \frac{7}{3} - \frac{1}{e}.$$

4. 设  $f(x)$  是连续函数, 证明

$$(1) \int_0^a x^3 f(x^2)dx = \frac{1}{2} \int_0^{a^2} xf(x)dx.$$

$$\text{分析: } \int_0^a x^3 f(x^2)dx = \frac{1}{2} \int_0^a x^2 f(x^2)dx^2.$$

证明: 令  $t = x^2$ , 则  $x = \sqrt{t}$ ,

$$dx = \frac{1}{2\sqrt{t}} dt.$$

当  $x=0$  时,  $t=0$ ; 当  $x=a$  时,  $t=a^2$ .

$$\text{于是 } \int_0^a x^3 f(x^2)dx = \frac{1}{2} \int_0^a x^2 f(x^2)dx^2$$

$$= \frac{1}{2} \int_0^{a^2} tf(t)dt = \frac{1}{2} \int_0^{a^2} xf(x)dx.$$

$$(2) \int_0^{\pi} f(|\cos x|)dx = 2 \int_0^{\frac{\pi}{2}} f(|\cos x|)dx .$$

$$\text{证明: } \int_0^{\pi} f(|\cos x|)dx$$

$$= \int_0^{\frac{\pi}{2}} f(|\cos x|)dx + \int_{\frac{\pi}{2}}^{\pi} f(|\cos x|)dx .$$

$$\text{对积分 } \int_{\frac{\pi}{2}}^{\pi} f(|\cos x|)dx ,$$

$$\text{令 } t = \pi - x , \text{ 则 } dx = -dt .$$

$$\text{当 } x = \frac{\pi}{2} \text{ 时, } t = \frac{\pi}{2}; \text{ 当 } x = \pi \text{ 时, } t = 0 .$$

于是

$$\int_{\frac{\pi}{2}}^{\pi} f(|\cos x|)dx = \int_{\frac{\pi}{2}}^0 f(|\cos(\pi - t)|)d(-t)$$

$$= \int_0^{\frac{\pi}{2}} f(|-\cos t|)dt = \int_0^{\frac{\pi}{2}} f(|\cos t|)dt$$

$$= \int_0^{\frac{\pi}{2}} f(|\cos x|)dx , \text{ 所以}$$

$$\int_0^{\pi} f(|\cos x|)dx = 2 \int_0^{\frac{\pi}{2}} f(|\cos x|)dx .$$

#### § 5—4

##### 1. 选择题

(1) 以下反常积分收敛的是 ( C ).

$$(A) \int_1^{+\infty} \frac{1}{\sqrt{x}} dx; \quad (B) \int_1^{+\infty} dx;$$

$$(C) \int_0^1 \frac{1}{\sqrt{x}} dx; \quad (D) \int_0^1 \frac{1}{x^2} dx .$$

(2) 对反常积分  $\int_2^{+\infty} f(x)dx$  的收敛性,

正确的是 ( B ).

(A) 对任何  $p$  发散; (B)  $p < 1$  发散;

(C)  $p = 1$  收敛; (D)  $p > 1$  发散.

2. 判别下列反常积分的敛散性,

如果收敛, 计算它的值

$$(1) \int_1^e \frac{dx}{x\sqrt{1-(\ln x)^2}}.$$

$$\text{解: 原式} = \int_1^e \frac{d(\ln x)}{\sqrt{1-(\ln x)^2}}$$

$$= [\arcsin(\ln x)]_1^e$$

$$= \lim_{x \rightarrow e^-} \arcsin(\ln x) - \arcsin(\ln 1)$$

$$= \frac{\pi}{2} - 0 = \frac{\pi}{2}.$$

这反常积分收敛.

$$(2) \int_0^2 \frac{x^2}{x^3-1} dx.$$

$$\text{解: 因为 } \lim_{x \rightarrow 1} \frac{x^2}{x^3-1} = \infty,$$

所以  $x=1$  为被积函数的瑕点.

又因为

$$\begin{aligned} \int_1^2 \frac{x^2}{x^3-1} dx &= \frac{1}{3} \int_1^2 \frac{1}{x^3-1} d(x^3-1) \\ &= \frac{1}{3} [\ln(x^3-1)]_1^2 \\ &= \frac{1}{3} [\ln(8-1) - \lim_{x \rightarrow 1^+} \ln(x^3-1)] = +\infty, \end{aligned}$$

则反常积分  $\int_1^2 \frac{x^2}{x^3-1} dx$  发散,

所以原反常积分  $\int_0^2 \frac{x^2}{x^3-1} dx$  发散.

$$(3) \int_1^{+\infty} \frac{\ln x}{x^2} dx.$$

$$\text{解: } \int_1^{+\infty} \frac{\ln x}{x^2} dx = - \int_1^{+\infty} \ln x d\left(\frac{1}{x}\right)$$

$$= - \left[ \frac{1}{x} \ln x \right]_1^{+\infty} + \int_1^{+\infty} \frac{1}{x} d(\ln x)$$

$$= -\left(\lim_{x \rightarrow +\infty} \frac{\ln x}{x} - 0\right) + \int_1^{+\infty} \frac{1}{x^2} dx$$

(洛必达法则)

$$= -\lim_{x \rightarrow +\infty} \frac{1}{x} - \left[\frac{1}{x}\right]_1^{+\infty} = 0 - \left(\lim_{x \rightarrow +\infty} \frac{1}{x} - 1\right)$$

$$= -(0 - 1) = 1.$$

3. 当  $k$  为何值时,

广义积分  $\int_2^{+\infty} \frac{dx}{x(\ln x)^k}$  收敛?

解: 当  $k=1$  时,

$$\int_2^{+\infty} \frac{dx}{x(\ln x)^k} = \int_2^{+\infty} \frac{dx}{x(\ln x)}$$

$$= \int_2^{+\infty} \frac{d(\ln x)}{\ln x} = [\ln(\ln x)]_2^{+\infty}$$

$$= \lim_{x \rightarrow +\infty} \ln(\ln x) - \ln(\ln 2) = +\infty,$$

这广义积分发散.

当  $k \neq 1$  时,

$$\int_2^{+\infty} \frac{dx}{x(\ln x)^k} = \int_2^{+\infty} (\ln x)^{-k} d(\ln x)$$

$$= \left[ \frac{1}{1-k} (\ln x)^{1-k} \right]_2^{+\infty}$$

$$= \frac{1}{1-k} \left[ \lim_{x \rightarrow +\infty} (\ln x)^{1-k} - (\ln 2)^{1-k} \right]$$

$$= \begin{cases} +\infty, & \text{当 } k < 1 \text{ 时,} \\ \frac{(\ln 2)^{1-k}}{k-1}, & \text{当 } k > 1 \text{ 时.} \end{cases}$$

故当  $k > 1$  时, 这广义积分收敛于

$$\frac{(\ln 2)^{1-k}}{k-1}; \text{ 当 } k \leq 1 \text{ 时, 这广义积分发散.}$$